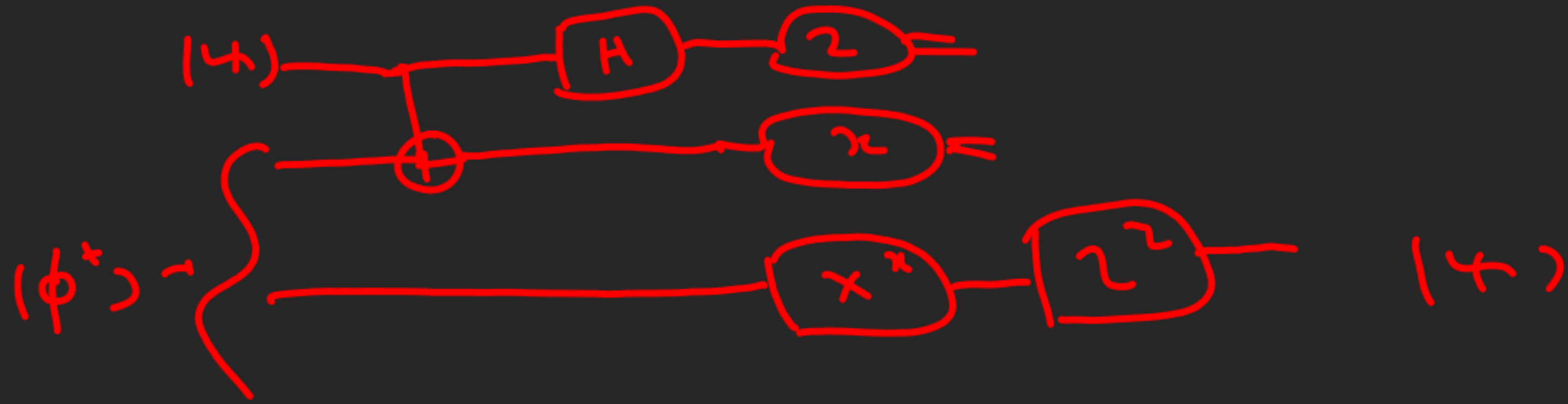
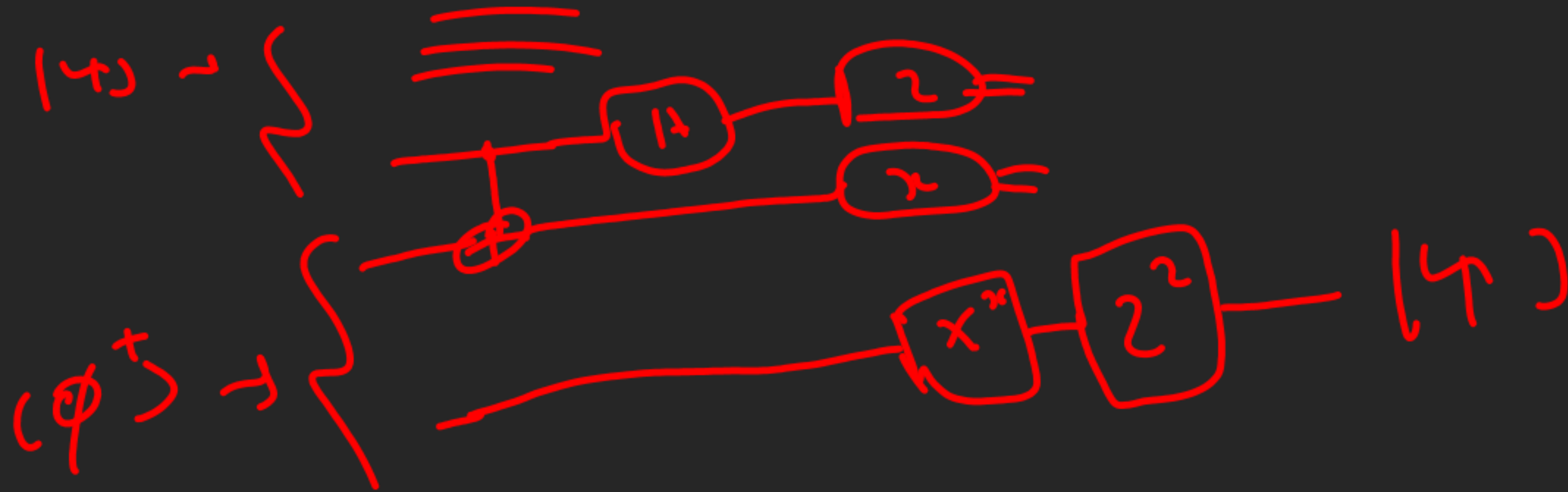


Quantum Teleportation



$$|4\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$



Also works
for multi-qubit
system

Superdense Coding

→ we can send 2 classical using 1 qubit and a shared pair of ebit

Alice



Bob

x_2	Alice's action	Bob's action
00	$ \phi^+\rangle \rightarrow \phi^+\rangle$	$ \phi^+\rangle \rightarrow +\rangle 0\rangle \rightarrow 00\rangle$
01	$ \phi^+\rangle \rightarrow \phi^-\rangle$	$ \phi^-\rangle \rightarrow -\rangle 0\rangle \rightarrow 10\rangle$
10	$ \phi^+\rangle \rightarrow \psi^+\rangle$	$ \psi^+\rangle \rightarrow +\rangle 1\rangle \rightarrow 01\rangle$
11	$ \phi^+\rangle \rightarrow \psi^-\rangle$	$ \psi^-\rangle \rightarrow -\rangle 1\rangle \rightarrow 11\rangle$

1 qubit $\leftarrow$ ^{1 ebit} 2 cbits